

X-ray photoelectron spectroscopy (XPS) of Quinacridone (QA) on the Ag(100) surface

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October 7, 2025

Motivation

- QA shows 1D molecular chains due to intermolecular hydrogen bonds^[1-4]
- Previous studies revealed 2 distinct phases of QA on Ag(100)
- **But:** Molecular interactions & energetics of phase transition are not fully understood

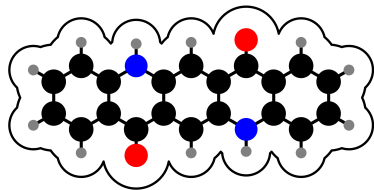


Figure: Ball-and-stick model of QA:
C: black, H: grey, N: blue, O: red.

Research Question

How do intermolecular hydrogen bonds & adsorbate-substrate interactions change during phase transitions?

Known α -Phase of QA on Ag(100)

α -Phase:

- Forms at 300 K
- point-on-line (POL) structure
- Homochiral 1D chains
- 2 hydrogen bonds per molecule
- Metastable phase

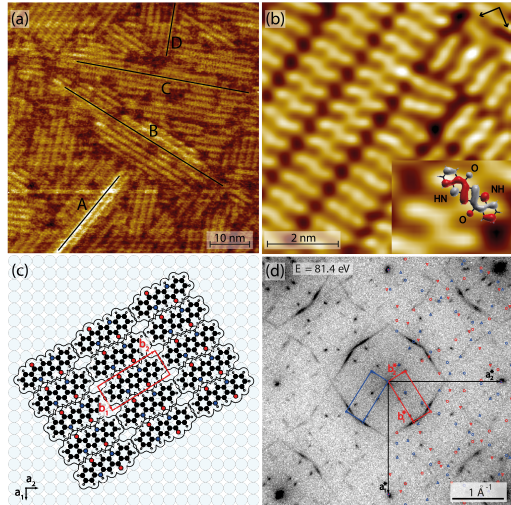


Figure: α -phase of QA on Ag(100).^[1]

Known β -Phase of QA on Ag(100)

β -Phase:

- Forms after heating to 500 K
- Commensurate structure
- 2D network with dimers
- Proposed 1.5 H-bonds per molecule
- Thermodynamically stable

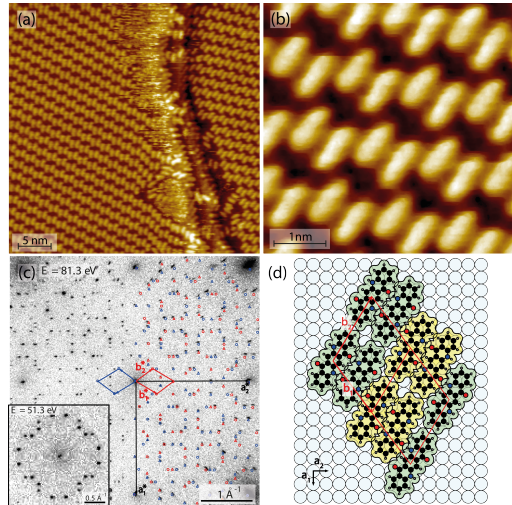


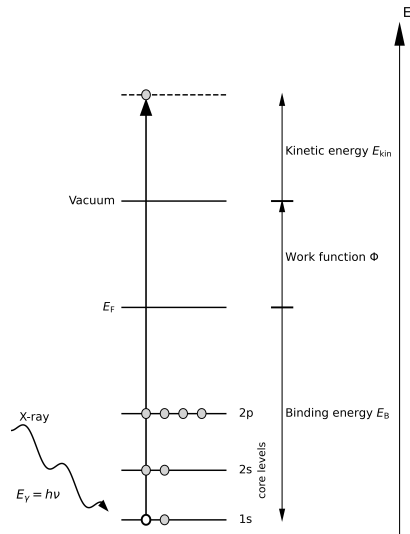
Figure: β -phase of QA on Ag(100).^[1]

X-ray photoelectron spectroscopy

$$E_{\text{kin}} = h\nu - E_B - \phi_{\text{sample}}$$

Key Features:

- Surface-sensitive ($\sim$ nm depth)
- Element-specific core levels
- Binding energy reflects local environment



Experimental Setup

Location: Diamond Light Source (I09 beamline), UK

- UHV chamber ($p \approx 10^{-10}$ mbar)
- Synchrotron radiation source
- Variable photon energy



Figure: Diamond Light Source, UK.^[5]

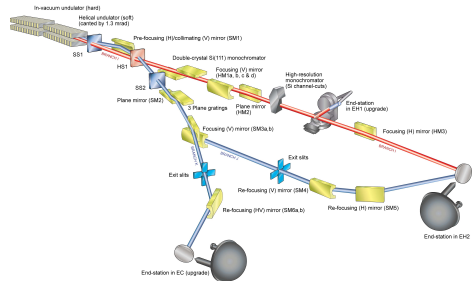


Figure: I09 endstation schematic.^[5]

C1s XPS Spectra

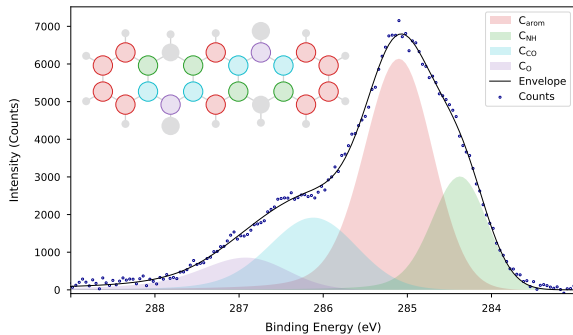


Figure: C1s spectrum of α -phase.

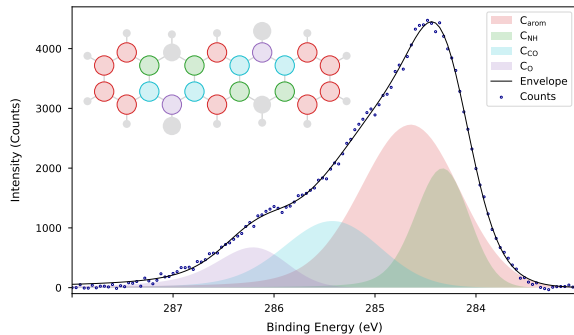


Figure: C1s spectrum of β -phase.

N1s XPS Spectra

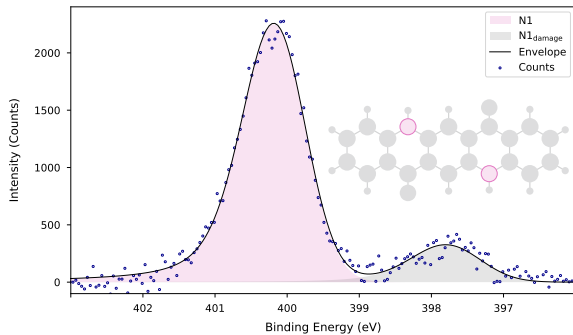


Figure: N1s spectrum of α -phase.

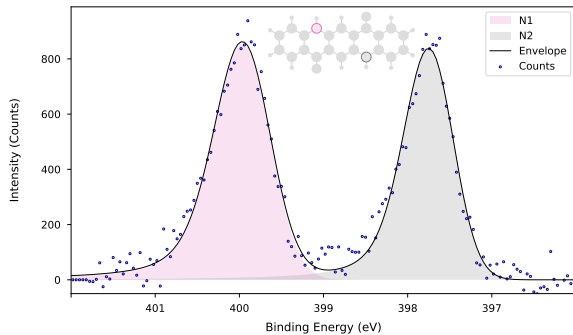


Figure: N1s spectrum of β -phase.

O1s XPS Spectra

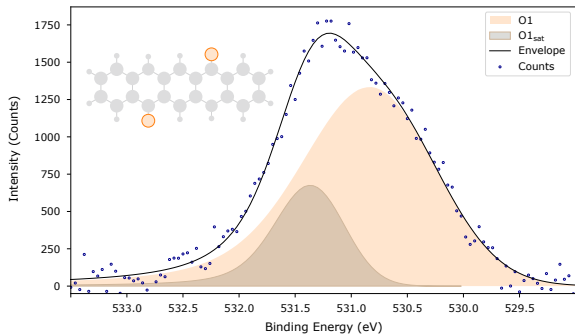


Figure: O1s spectrum of α -phase.

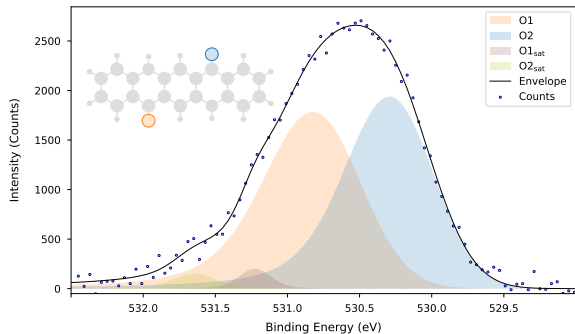
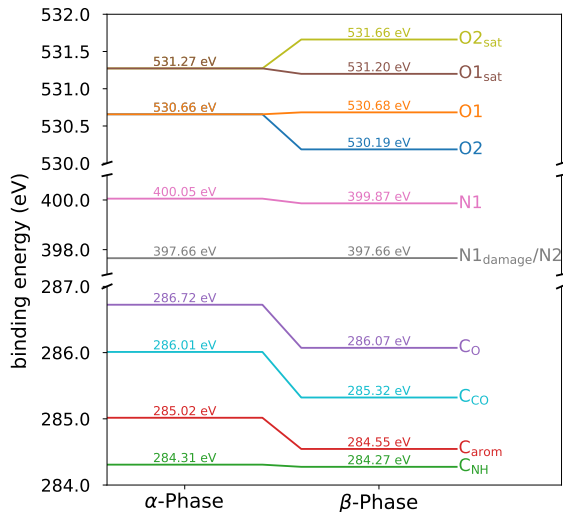


Figure: O1s spectrum of β -phase.

Binding Energy Changes



- **General trend:** Lower BEs in β -phase
→ stronger surface interaction
- **C1s:** Uniform shifts except for C_{NH}
- **N1s:** Large 2.2 eV shift for N2 component
- **O1s:** 0.5 eV difference between O1 and O2

Figure: Binding energy shifts from α - to β -phase.

Deprotonation Evidence

N1s Shift:

- 2.2 eV shift matches literature for deprotonated amine group^[6–8]
- Similar to beam damage in α -phase

Supporting Evidence:

- O1s shows 2 distinct environments
- One O maintains H-bonding (O1)
- Other O loses H-bonding partner (O2)

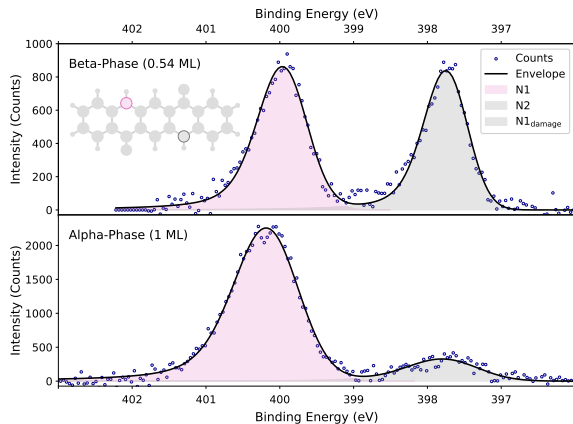


Figure: N1s XPS spectra for α - & β -phase.

Key Findings

α -Phase

- Single peaks for N and O atoms $\rightarrow$ equivalent environments
- Confirms 2 hydrogen bonds per molecule

β -Phase

- 2 distinct N & 2 distinct O environments (1:1 area ratio)
- Large N1s shift suggests deprotonation of 1 amine group
- Evidence for only 1 hydrogen bond per molecule

New Interpretation

Phase transition involves deprotonation of 1 amine group per molecule!

Structure Models: α - vs. β -Phase

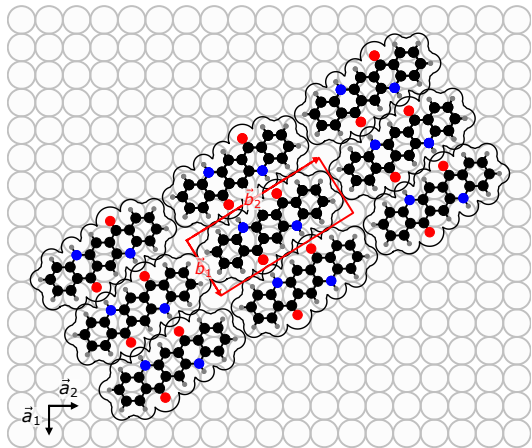


Figure: α -phase: 2 H-bonds per molecule, all N and O atoms equivalent

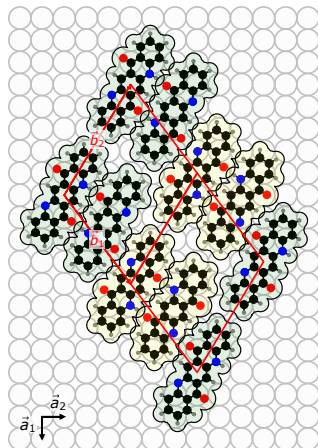


Figure: β -phase: 1 H-bond per molecule, one N deprotonated

Summary of Main Results

1 Confirmed α -phase structure:

- 2 hydrogen bonds per molecule
- All N & all O atoms have equivalent environments
- Consistent with existing structure model

2 Revised β -phase understanding:

- 1 hydrogen bond per molecule (not 1.5)
- 1 deprotonated amine group per molecule, 2 distinct N & 2 distinct O environments
- Requires new structure model

3 Phase transition:

- Deprotonation of amine group
- Stronger molecule-surface interactions in β -phase

Key Impact

New insights into molecular interactions and phase behavior of QA on the Ag(100) surface, including evidence for deprotonation during phase transition.

Follow-up Studies

- **Normal incidence X-ray standing wave (NIXSW):** Determine exact adsorption heights and sites
- **Mass spectrometry:** Detect H_2 evolution during phase transition
- **Other derivatives:** Study similar systems, e.g. FQA

References I

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- [2] Thorsten Wagner et al. "Quinacridone on Ag(111): Hydrogen Bonding versus Chirality". In: *The Journal of Physical Chemistry C* 118.20 (May 22, 2014), pp. 10911–10920.
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- [4] Alexander Richard Eberle. "Growing Low-Dimensional Supramolecular Crystals via Organic Solid-Solid Wetting Deposition". Ludwig-Maximilians-Universität München, Feb. 2019.
- [5] Diamond. *Diamond Light Source: I09*. June 2025. URL: <https://www.diamond.ac.uk/Instruments/Structures-and-Surfaces/I09.html>.
- [6] G. Ruano et al. "Deprotonation of the Amine Group of Glyphosate Studied by XPS and DFT". In: *Applied Surface Science* 567 (Nov. 30, 2021), p. 150753.
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- [8] Muzamir M. Mahat et al. "Elucidating the Deprotonation of Polyaniline Films by X-ray Photoelectron Spectroscopy". In: *Journal of Materials Chemistry C* 3.27 (2015), pp. 7180–7186.
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- [13] Hongxiang Xu et al. "On-Surface Isomerization of Indigo within 1D Coordination Polymers". In: *Angewandte Chemie* 136.15 (2024), e202319162.

Thank you!

Thank you for your attention!

Questions?

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Quinacridone Properties

Chemical Properties:

- Formula: $C_{20}H_{12}N_2O_2$
- Prochiral organic molecule
- Highly conjugated π -system
- Symmetry group: C_{2h}

Applications:

- Pigment in industry (Violet 19)
- Well-defined surface structures
- Organic electronics components

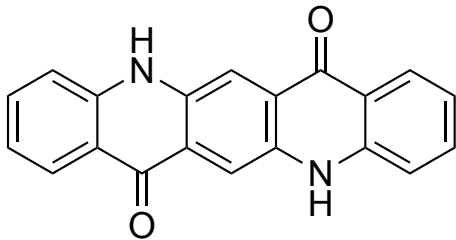
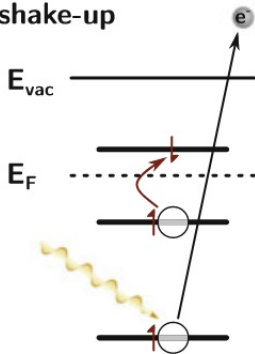


Figure: Structure of QA.

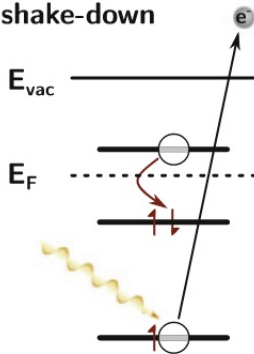
Multielectron processes in XPS

Multi electron processes

(c) shake-up



(d) shake-down



(e) shake-off

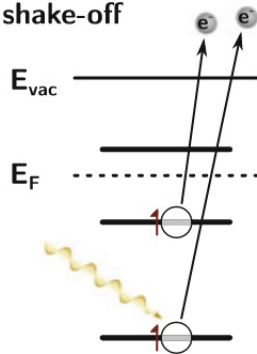


Figure: Multielectron processes in XPS. Taken from [9].

Table: Comparison of experimental results with literature values.

System	C _{arom} BE (eV)	N1s BE (eV)	O1s BE (eV)
QA/ α -phase	285.02	400.05	530.66
QA/ β -phase	284.55	399.87/397.66	530.68/530.19
PTCDA ^[10]	284.91	-	531.93
Indigo ^[11-13]	285.5	399.9	530.2

- Good agreement with similar aromatic systems
- Large N1s shift consistent with deprotonation studies
- O1s values typical for carbonyl groups
- Phase transition shifts indicate stronger surface coupling

LEED Patterns

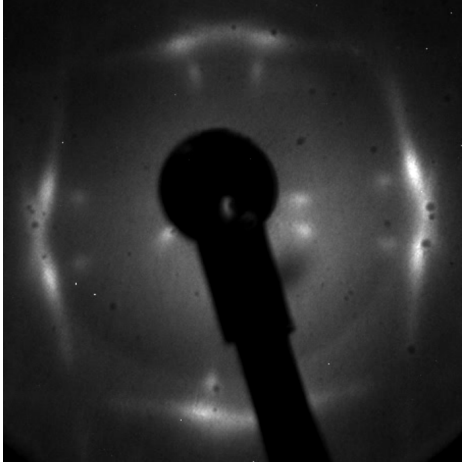


Figure: Recorded LEED pattern of α -phase at DLS.

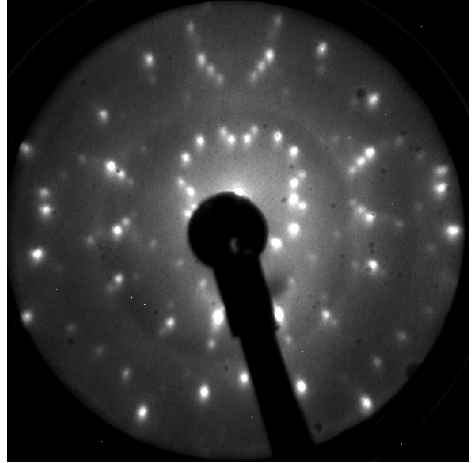


Figure: Recorded LEED pattern of β -phase at DLS.

Superstructure of QA on Ag(100)

- POL structure for α -phase^[1]
→ every point of superstructure falls on substrate lattice line in [10] direction

$$\mathbf{M}_{\alpha} = \begin{pmatrix} 2 & 1.25 \\ -3 & 4.80 \end{pmatrix}$$

- $b_1 = (6.8 \pm 0.1) \text{ \AA}$
- $b_2 = (16.4 \pm 0.1) \text{ \AA}$
- $\alpha = 90^\circ \pm 1^\circ$

- Commensurate structure for β -phase^[1]

$$\mathbf{M}_{\beta} = \begin{pmatrix} 4 & 3 \\ -5 & 3 \end{pmatrix}$$

- $b_1 = (14.4 \pm 0.1) \text{ \AA}$
- $b_2 = (16.8 \pm 0.1) \text{ \AA}$
- $\alpha = 112^\circ \pm 1^\circ$

C1s spectra

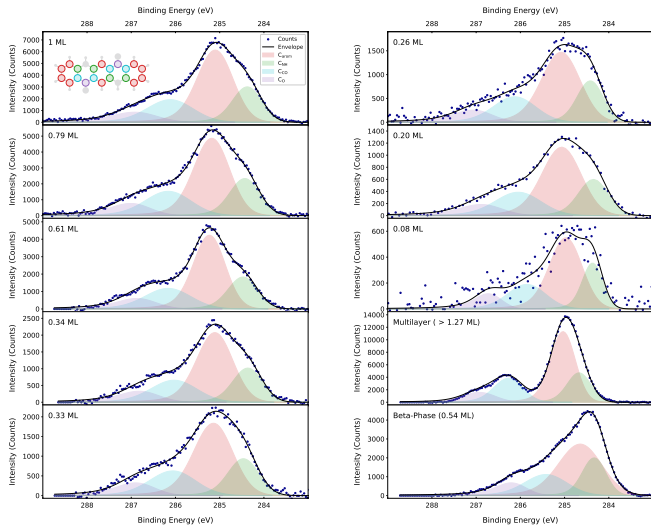


Figure: Overview of C1s XPS spectra for QA on Ag(100).

N1s spectra

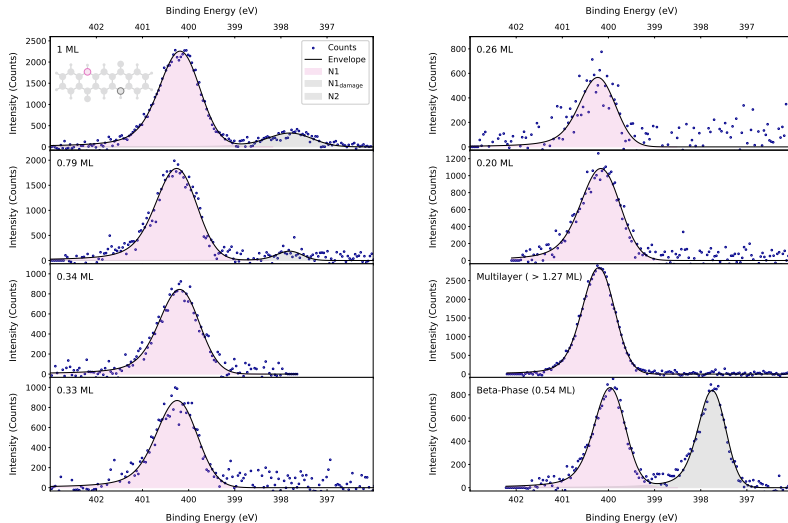


Figure: Overview of N1s XPS spectra for QA on Ag(100).

O1s spectra

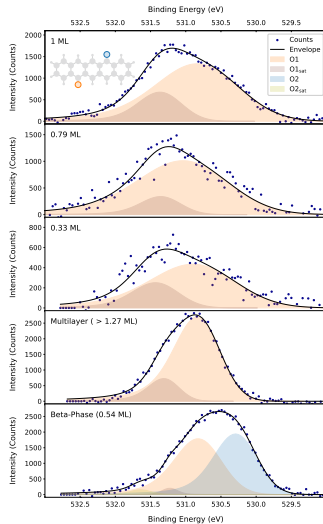


Figure: Overview of O1s XPS spectra for QA on Ag(100).

XPS fitting parameter

peak	line shape	area ratio
C _{arom}	LA(1,8,800)	10
C _{NH}	LA(1,8,800)	4
C _{CO}	LA(1,8,800)	4
C _O	LA(1,8,200)	2
O1	LA(1,8,400)	1 (for β -phase)
O1 _{sat}	LA(1,1,400)	-
O2	LA(1,8,400)	1 (for β -phase)
O2 _{sat}	LA(1,1,400)	-
N1	LA(1,8,400)	1 (for β -phase)
N1 _{damage}	LA(1,8,400)	-
N2	LA(1,8,400)	1 (for β -phase)

Parameters of XPS fits

Phase	Core-level	Component	BE (eV)	FWHM (eV)	Area
α	C1s	C _{arom}	285.02	0.94	10
		C _{NH}	284.31	0.76	4
		C _{CO}	286.01	1.20	4
		C _O	286.72	1.15	2
	N1s	N1	400.05	1.02	-
		N1 _{damage}	397.66	1.16	-
	O1s	O1	530.66	1.36	-
		O1 _{sat}	531.27	0.71	-
β	C1s	C _{arom}	284.55	1.13	10
		C _{NH}	284.28	0.62	4
		C _{CO}	285.32	1.11	4
		C _O	286.07	0.79	2
	N1s	N1	399.87	1.20	1
		N2	397.66	1.20	1
	O1s	O1	530.68	0.71	1
		O1 _{sat}	531.20	0.45	-
		O2	530.19	0.68	1
		O2 _{sat}	531.66	0.36	-